The Saturday Mornings.tv

TODAY'S FORECAST
MOSTLY SUNNY,
SCATTERED ADB BY LUNCH

THE FREE EDITION
ANDROID TV MANUAL
BACK TO THE FRONT PAGE

DO-IT-YOURSELF SUPPLEMENT · PART THREE SATURDAYMORNINGS.TV

"Please adjust your set"

THE FREE EDITION · ANDROID LANE

ANDROID TV, STEP BY STEP

How to put the Saturday Mornings TV app on your own Android TV or Google TV: seven taps to become a developer, Android's free tools, one settings file and one script. No certificates, no accounts.

9 STEPS | 0 CERTIFICATES | 1 REMOTE CONTROL | ANDROID & GOOGLE TV

Difficulty, in bowls of cereal: 🥣 easy · 🥣🥣 fiddly · 🥣🥣🥣 pour yourself another bowl

FIRST: BUILD YOUR STATION SAMSUNG TV GUIDE THE SHORT VERSION FRONT PAGE

WHAT YOU'LL NEED

- [ ] **A working station**: its address answers `{"ok":true}` at `/api/health`. Not yet? Build it first.
- [ ] **An Android TV or Google TV**: an NVIDIA Shield, a Chromecast with Google TV, or a Sony, TCL, Hisense or Philips set running Google TV. Tested on a Shield.
- [ ] **A computer** with a terminal. The build script is written for Mac and Linux.
- [ ] **The same Wi-Fi** for the TV and the computer.
- [ ] **Java 17 and Android's command-line tools**, free (about a gigabyte), or Android Studio, which includes them.

☐ **The code** from github.com/RonzGore/saturday-mornings, already on your computer from the station guide.

TIME, FIRST GO	MOSTLY SPENT	COST	EACH UPDATE AFTER
about 45 minutes	**downloading**	**$0**	**2 minutes**

PART 1 · THE TOOLS — *Android's toolkit on your computer*

1

Install Java and Android's tools

The app is built with Android's own free tools, and they need Java 17. On a Mac with Homebrew, three lines do it (on Linux, install Java 17 and Android's command-line tools from your package manager or `developer.android.com`, or install Android Studio):

TYPE THIS (MAC)

```
brew install openjdk@17
brew install --cask android-commandlinetools android-platform-tools
sdkmanager --sdk_root=$HOME/Library/Android/sdk "platforms;android-36" "build-tools;36.0.0"
```

It asks you to accept Android's licences; type `y`.

> **YOU SHOULD SEE:** `adb version` prints a version number. (`adb` is the tool that talks to the TV.)

20 MIN

PART 2 · THE TV — *Seven taps and a handshake*

2

SECRET!

Become a developer

On the TV, open **Settings › Device Preferences › About** (on Google TV: **Settings › System › About**). Scroll to **Build** (or **Android TV OS build**) and press OK on it **seven times**.

> **YOU SHOULD SEE:** "You are now a developer", and a new **DEVELOPER OPTIONS** entry under Device Preferences (or System).

2 MIN

3

Turn on network debugging

In **Developer options**, turn on **USB debugging**, then **Network debugging** (on a Shield). On newer Google TVs it's called **Wireless debugging**. Turn it on, choose **Pair device with pairing code**, and type the pairing line it shows on your computer:

GOOGLE TV ONLY: TYPE THIS, WITH THE NUMBERS YOUR TV SHOWS

```
adb pair 192.168.1.60:37000 123456
```

YOU SHOULD SEE: Network or Wireless debugging switched on, with the TV's address next to it.

3 MIN

4

Find the TV's address

Usually shown right under Network debugging, like `192.168.1.60:5555`. If not: **Settings** › **Network & Internet**, then your Wi-Fi network.

YOU SHOULD SEE: an address like `192.168.1.60`. Write it down.

2 MIN

PART 3 · THE APP *Point it, build it, send it*

5

Point the app at your station

Everything for the Android app lives in `apps/android-tv`. Your own settings go in a small file there called `local.properties` (it stays on your computer, since git ignores it). It holds where Android's tools are, your station's address ending in `/tv`, your TV's address, and your own app id.

TYPE THIS, FROM THE SATURDAY-MORNINGS FOLDER (YOUR OWN ADDRESSES)

```
printf 'sdk.dir=%s\ntv.server=https://tv.your-
domain.com/tv\ntv.ip=192.168.1.60\ntv.appId=com.yourname.saturdaymornings\n'
"$HOME/Library/Android/sdk" > apps/android-tv/local.properties
```

No domain? Use `https://saturday-morning.YOUR-SUBDOMAIN.workers.dev/tv`.

YOU SHOULD SEE: the TV app when you open that exact station address in a browser on your computer.

5 MIN

6

Build and install

One script builds the app, connects to the TV, installs it and opens it. The very first build downloads Android's build tool (Gradle), so it takes a few minutes.

TYPE THIS, FROM THE SATURDAY-MORNINGS FOLDER

```
apps/android-tv/build.sh
```

The first time, the TV asks **"Allow network debugging?"**. Tick **Always allow from this computer** and press OK. Missed it? Run the script again.

> **YOU SHOULD SEE:** `Success`, then `opened on 192.168.1.60:5555`, and Saturday Mornings full screen on the TV.

10 MIN

PART 4 · ON THE AIR *Pair up, sit back*

7

Link it from your phone

The TV shows a pairing code and a QR code. In the parent app on your phone, go to **Your TVs** › **+ Link a TV** and type the code, or scan the QR.

> **YOU SHOULD SEE:** the TV switch to your kids' channels. From now on it checks for your changes every 30 seconds.

2 MIN

8

Find it again

Saturday Mornings sits in the TV's **Apps** row with its own banner (on some Google TVs, behind **See all**). Hold OK on it to move it to the front.

> **YOU SHOULD SEE:** the Saturday Mornings banner next to your other apps.

1 MIN

9

Updating the TV app later

The TV screens come from your station, so most updates reach the TV by themselves. Only when something in `apps/android-tv` changes do you reinstall: TV on, network debugging still on, same script. Want it on Google Play? See the release steps in `docs/SELF-HOSTING.md`.

TYPE THIS

```
apps/android-tv/build.sh
```

YOU SHOULD SEE: the same `Success` as step 6. Your TV stays linked.

2 MIN

STUCK?

`unauthorized` **or** `failed to authenticate`	The TV is waiting for you to press OK on "Allow network debugging?". No box? Turn Network debugging off and on again in Developer options, then run the script again.
`failed to connect` **or it times out**	TV switched on and awake (not asleep with the screen off)? Same Wi-Fi? Home routers sometimes hand out new addresses, so check step 4 and update `tv.ip`.
"Can't reach the station"	It's usually the `tv.server` address in `local.properties`, with a typo, a missing `https://` or `/tv`, or a station that isn't up (check `/api/health`). Fix it and run the script again. The app also retries by itself, so a TV that was just waking up sorts itself out.
The app closed by itself	When Android updates its built-in web engine (Android System WebView), it closes every app using it. Just open it again; a TV that was being linked picks the link up.
`SDK location not found`**, or Java errors**	Check `sdk.dir` in `local.properties` points at Android's tools, and that Java 17 is installed (the script uses Homebrew's on a Mac).
The pairing code won't link	Make sure the phone is signed in to the same station the TV points at. Type the code while it's still on screen.

Still stuck? Open an issue on the code's GitHub page with the command you ran and exactly what it said.

That's a TV. *Grab the cereal, sit the kids down, and* **enjoy your Saturday mornings.** *Haven't built your station yet? Start there. Got a Samsung too? Here's its guide. Rather not do any of this? The home-delivery edition installs in three steps.*

